

Live4D: A Real-time Capture System for Streamable Volumetric Video

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Figure 1: Stream-ready meshes with corresponding texture maps generated by the Live4D system and some application scenarios, such as 3D holographic displays and virtual reality environments. These media assets can be distributed to different clients for real-time decoding and rendering.

ABSTRACT

Volumetric video holds promise for virtual and augmented reality (VR/AR) applications but faces challenges in interactive scenarios due to high hardware costs, complex processing and substantial data

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streams. In this paper, we introduce Live4D, a cost-effective, real-time volumetric video generation and streaming system using an RGB-only camera setup. We propose a novel deep implicit surface reconstruction algorithm, that combined neural signed distance field with observed truncated signed distance field to generate the watertight meshes with low latency. Moreover, we achieve a robust non-rigid tracking method that provides temporal stability to the meshes while resisting tracking failure cases. Experimental results show that Live4D achieves a performance of 24fps using mid-range graphic cards and exhibits an end-to-end latency of 95ms. The system enables live streaming of volumetric video within a 20Mbps bandwidth requirement, positioning Live4D as a promising solution for real-time 3D vision content creation in the growing VR/AR industry.

CCS CONCEPTS

• **Human-centered computing** → **Mixed / augmented reality**;
 • **Computing methodologies** → **Reconstruction**; **Image processing**; **Computer vision**; **Computer graphics**; • **Computer systems organization** → **Real-time systems**.

KEYWORDS

Real-time, Non-Rigid Reconstruction, Volumetric Capture

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1 INTRODUCTION

The innovative technology of volumetric video offers users enhanced degrees of freedom, enabling immersive experiences by directly transferring surface meshes and texture maps to the client display. This technology has gained significant momentum in response to the emergence of consumer virtual reality (VR) and augmented reality (AR) hardware, signaling its potential to revolutionize visual experiences for the general public in ways that surpass the capabilities of conventional video.

Recording high-quality geometry and texture typically necessitates a significant number of sensors with densely sampling the view port. The remarkable research [Collet et al. 2015] showcases a state-of-the-art capture system for producing high-quality streamable volumetric video, highlighting its impressive potential for free-viewpoint videos. However, these systems tend to be complex, incur high production costs, and involve intricate workflows, limiting their practicality in streamable and interactive applications.

In this paper, our focus lies on consumer scenarios, where we present a cost-effective and robust end-to-end solution for volumetric video capture.

2 APPROACH

The entire processing pipeline is depicted in Figure 2. Our system comprises two key components: the front end and the reconstruction server. The front end includes flexible number of calibrated binocular industrial cameras and a computational station responsible for processing raw data and generating multi-view synchronized RGB-D frames. Multiple image-based tasks, including detection, background matting and stereo matching, are subsequently applied to the RGB data. The resulting RGB-D data and additional information are then transmitted to the reconstruction server for further processing. The reconstruction phase utilizes a hybrid method combining TSDF fusion and neural implicit surface reconstruction to achieve watertight implicit surface reconstruction. Furthermore, the complete TSDF volume is enhanced by fusing historical geometry information based on non-rigid tracking, resulting in temporally stable meshes. Subsequently, several mesh processing techniques are used to recover finely detailed geometry with streamable topology

and temporal coherently mapped atlas, ensuring efficient compression within the available bandwidth. The final data is sent to the user clients for further applications.

2.1 Depth Acquisition

Though deep learning based method could acquire finer depth information, none of these methods are capable of obtaining competitive results in the time cost we required. Thus, modified from the real-time model of [Lipson et al. 2021], we construct the whole algorithm from scratch combined TensorRT¹ with custom CUDA² operation. Moreover, we developed a region of interest strategy that focuses on improving the accuracy of disparities in the important areas like faces. The strategy involves identifying and cropping the regions of interest from the input images and separately processing them in the stereo matching module. For better performance, we distill the model, using light-weight matting and confidence network to filter the outliers.

2.2 Implicit Surface Reconstruction

Previous work such as [Li et al. 2020; Yu et al. 2021] has leveraged a learning-based method to train an implicit function for real-time 3D shape reconstruction. Considering our heavy system with limited computational budgets, these method can still be computationally expensive. In our work, we address this challenge by reusing truncated signed distance function (TSDF)[Curless and Levoy 1996], combined with a neural SDF inferred by an implicit function, which enables the generation of watertight meshes with highly computational efficiency.

We utilize the underlying TSDF volume as the surface representation, fusing multi-view depth data per frame. Considering memory footprint and computational cost for representing fine-geometry with dense voxel grids, our proposed approach involves a sparse multi-resolution volume structure, integrated with implicit neural SDF for implicit completion.

Training. Based on the encoder-decoder architecture described in [Park et al. 2019], we training the implicit function (IF) to estimate the neural SDF field and clamp the output of the IF to the range of $(-1, 1)$, approximating the TSDF field. This strategy encourages the network to estimate the signed distance accurately near the surface, while classifying the occupancy of voxels far from the surface. Denoting MLP as $f(\cdot)$, we directly use fused TSDF volume combined with blurred visual hull volume $\mathcal{X} \subset \mathbb{R}^{2 \times H \times W \times D}$ as our input of encoder $G(\cdot)$ and perform an end-to-end training to predict more smooth TSDF values $\hat{x} = f(G(\mathcal{X}), x) = F(x, p)$ and complete the TSDF field for unseen regions. For each point, the observed TSDF value x as an extra hand-crafted feature is concatenated to preserve geometry details and better fit the observed TSDF field. Objective functions are formulated as follows:

$$\mathcal{L} = \mathcal{L}_{recon} + \lambda_{TV} \mathcal{L}_{TV} \quad (1)$$

$$\mathcal{L}_{recon} = \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} \|\hat{x} - x^*\| + \lambda_{consis} \frac{1}{|\mathcal{P}'|} \sum_{p \in \mathcal{P}'} \rho(\|\hat{x} - x\|) \quad (2)$$

¹<https://developer.nvidia.com/tensorrt>

²<https://developer.nvidia.com/cuda-toolkit>

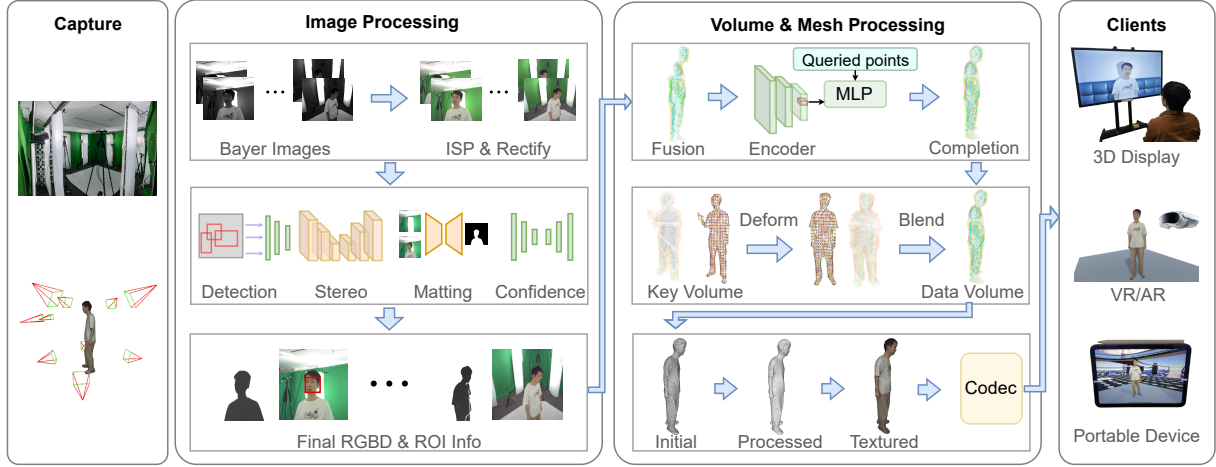


Figure 2: Overview of the Live4D system.

Given random sampled points $\mathcal{P}$, $\mathcal{L}_{recon}$ represents the reconstruction loss, x^* is the ground-truth (GT) TSDF value fused from dense view of GT depth maps. An extra term is proposed to keep consistent with observed TSDF values x of valid point set $\mathcal{P}' \subseteq \mathcal{P}$. $\rho(\cdot)$ denotes the Huber weight, allowing the imperfect observations. Total variation (TV) term $\mathcal{L}_{TV}$ [Rudin et al. 1992] is added to suppress the noise near-zero cross surfaces and encourage smoothness for the transition between observed TSDF field and the regions without observations.

Inference. The technique proposed by [Mescheder et al. 2019] introduced the use of octrees to accelerate inference in the context of learning-based implicit surface reconstruction. We take a step further to reconstruct highly detailed 3D surfaces in lower latency. We extend the existing techniques by incorporating the multi-scale hierarchy structure for partial inference scheduling and utilize observed information to direct the TSDF querying in the subsequent levels and skip voxels with observed TSDF value. This approach enables the completion of unobserved regions within the TSDF volume based on the information from observed regions, leading to a reduction in computational cost.

Non-rigid tracking. In volumetric video tasks, achieving temporal consistency is challenging due to the constantly changing topology and spatial appearance of objects over time. We implement the embedded deformation (ED) field [Sumner et al. 2007] based on dual-quaternion blending (DQB), as done in [Newcombe et al. 2015]. And the warp function is defined as $\mathcal{W}(\mathbf{p}) \equiv SE3(\mathbf{DQB}(\mathbf{p}))$. To solve the deformation field, we followed [Dou et al. 2016], and defined the energy function as $E(\mathcal{W}) = \lambda_{data}E_{data}(\mathcal{W}) + \lambda_{ARAP}E_{ARAP}(\mathcal{W})$. The data term E_{data} is a projective point-to-plane term that evaluates the misalignments between the deformed model and the observed data. Additionally, an as-rigid-as possible (ARAP) term E_{ARAP} is included as a regularization term that penalizes discontinuity of movements between all pairs of neighboring ED nodes. The deformation field is solved as a linear least squares (LLS) problem, which can be efficiently solved using the Levenberg-Marquardt (LM) solver in single GPU. To solve the large-scale sparse linear

equations in parallel, we implemented the preconditioned conjugate gradient (PCG) method, as proposed in [Dou et al. 2016]. Tracking failures resulting from inaccurate estimation of the deformation field can introduce noticeable artifacts during non-rigid fusion. To address this challenge, we propose a novel fusion strategy that combines the learnable prior with observed information to achieve higher robustness in the presence of tracking failures. Specifically, we directly evaluate per-voxel alignment error between the deformed volume and completed volume. The misaligned voxels are rejected and prevent from temporal fusion.

2.3 Geometry Processing

In the post-processing stage, we tackle the challenges to ensure efficient client-side streaming and rendering.

Surface extraction. We perform Marching Cubes [Lorensen and Cline 1987] to extract the surface based on the hierarchy structure. Given proposed multi-resolution voxel grid, neighboring voxels of differing resolutions generate the *crack artifacts* in isosurfaces. Leveraging an oriented edge table, we identify and handle crack edges across different resolutions, remove duplicate vertices from coarse voxels, reconnect the remaining vertices according to rules forming polygons. Then polygons is triangulated to patch cracks, yielding final isosurface.

Mesh simplification. To compress the geometry topology, based on [Wu and Kobbelt 2002], we proposed an optimization formula accelerating the decimation process by concurrently processing multiple edges during the collapse procedure. In line with this optimization approach, we construct a vertex-face adjacency list, which enables us to perform decimation steps within randomly sampled clusters of eight edges simultaneously. To handle potential collisions that may arise during this concurrent processing, we utilize a custom atomic operation implemented in CUDA.

Texturing. Texture recovery is a challenging task due to various factors such as inexact geometry, inaccurate calibration, and

Table 1: Geometry Reconstruction Results on RenderPeople

Method	Chamfer↓	P2S↓	Normal↓
PSR	1.958	1.604	0.337
IF-Net	4.527	4.208	0.475
Function4D	2.855	2.420	0.355
Ours	1.862	1.579	0.297

Table 2: Render Quality

Method	PSNR	SSIM	LPIPS
Function4D	28.88	0.7851	0.0315
Ours	30.51	0.8158	0.0217

color distortion in multi-view imaging, which often result in artifacts such as 'ghosting', seams, and image blurring. We generate a seamless texture map by leveraging weighted blending based on a distance field in the imaging space, approximating the geodesic field mentioned in [Du et al. 2018]. To address the challenge of missing texture information, which can significantly impact user-side applications and is difficult to obtain from prior knowledge, we propagate the colors from textured faces to untextured faces through multiple iterations via a half-edge mesh structure. Moreover, we enhance the propagation procedure by incorporating historical texture information through nonrigid fusion.

3 EXPERIMENTS

Performance. In our prototype capture setup, we utilize two capture station for capturing the RGB-D streams. Each station is equipped with five binocular cameras and two Nvidia RTX 3090 graphic cards which runs up to 25 FPS with an overall latency of approximately 50ms. The reconstruction computer with an Nvidia RTX 4090 graphic card and an Intel i9-12900K processor runs up to 28 FPS with roughly 45ms latency in total. For some clients, it takes 7ms for mesh decoding and 5.5ms for texture decoding in iPad mini 6th generation, while PICO Neo3 costs 4ms for mesh and 25ms for texture map. Some results are illustrated in Figure 1.

Surface reconstruction. In our evaluation, we compared three implicit surface reconstruction methods on Renderpeople³ Dataset: Poisson Surface Reconstruction (PSR) [Kazhdan and Hoppe 2013], Function4D [Yu et al. 2021], and IF-Net [Chibane et al. 2020] in Table 1. All three methods rely solely on depth inputs. Table 2 shows the render quality of Function4D and our method.

Streaming. We implement the mesh codec with NVIDIA Video SDK⁴ and Draco⁵. The mesh is initially generated with around 100k faces and 50k vertices from the hierarchy volume. After face decimation and point splitting for texturing purposes, we obtain around 20k faces and 18k vertices with uv attributes. A 1920×1080 texture maps is created for streaming purpose. We set the parameter of video compression with target quality as 28 and group of picture length as 8. We employ 11-bit quantization for vertex coordinates and 10-bit for uv coordinates. After encoding, the size of final mesh is around 65KB and texture map is around 40KB in average per frame.

³<https://renderpeople.com>

⁴<https://developer.nvidia.com/video-codec-sdk>

⁵<https://github.com/google/draco>

4 LIMITATIONS

Despite successfully improving the speed of deep learning-based stereo matching algorithms, our approach still requires substantial computational resources to obtain high-quality depth information. Furthermore, relying solely on RGB for stereo matching leads to inaccurate geometry in regions with low texture or pure black areas, directly impacting the reconstruction outcomes in the proposed Live4D system. Additionally, our method does not support transparent objects. Although we achieve a temporally stable implicit surface, the topology varies in the video sequence. Consequently, the consistency of the streamable meshes in the temporal domain is not sufficient. To enable streaming, we constrain the mesh and texture map image at an appropriate level, resulting in a decrease in the final rendering quality on VR/AR client devices. Furthermore, users can achieve more visually pleasing rendering results by adopting technologies like physically-based rendering to enhance the rendering outcomes.

REFERENCES

- Julian Chibane, Thimo Alldieck, and Gerard Pons-Moll. 2020. Implicit functions in feature space for 3d shape reconstruction and completion. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 6970–6981.
- Alvaro Collet, Ming Chuang, Pat Sweeney, Don Gillett, Dennis Evseev, David Calabrese, Hugues Hoppe, Adam Kirk, and Steve Sullivan. 2015. High-quality streamable free-viewpoint video. *ACM Transactions on Graphics* 34, 4 (2015), 1–13.
- Brian Curless and Marc Levoy. 1996. A volumetric method for building complex models from range images. In *Proceedings of the 23rd annual conference on Computer graphics and interactive techniques*. 303–312.
- Mingsong Dou, Sameh Khamis, Yuri Degtyarev, Philip Davidson, Sean Ryan Fanello, Adarsh Kowdle, Sergio Orts Escolano, Christoph Rhemann, David Kim, Jonathan Taylor, et al. 2016. Fusion4d: Real-time performance capture of challenging scenes. *ACM Transactions on Graphics* 35, 4 (2016), 1–13.
- Ruofei Du, Ming Chuang, Wayne Chang, Hugues Hoppe, and Amitabh Varshney. 2018. Montage4D: Interactive seamless fusion of multiview video textures. In *Proceedings of ACM Interactive 3D Graphics (I3D) 2018*.
- Michael Kazhdan and Hugues Hoppe. 2013. Screened Poisson Surface Reconstruction. *ACM Transactions on Graphics* 32, 3, Article 29 (jul 2013), 13 pages. <https://doi.org/10.1145/2487228.2487237>
- Ruilong Li, Yuliang Xiu, Shunsuke Saito, Zeng Huang, Kyle Olszewski, and Hao Li. 2020. Monocular Real-Time Volumetric Performance Capture. In *European Conference on Computer Vision*. Springer, 49–67.
- Lahav Lipson, Zachary Teed, and Jia Deng. 2021. Raft-stereo: Multilevel recurrent field transforms for stereo matching. In *2021 International Conference on 3D Vision (3DV)*. IEEE, 218–227.
- William E Lorensen and Harvey E Cline. 1987. Marching cubes: A high resolution 3D surface construction algorithm. *ACM siggraph computer graphics* 21, 4 (1987), 163–169.
- Lars Mescheder, Michael Oechsle, Michael Niemeyer, Sebastian Nowozin, and Andreas Geiger. 2019. Occupancy networks: Learning 3d reconstruction in function space. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 4460–4470.
- Richard A Newcombe, Dieter Fox, and Steven M Seitz. 2015. Dynamicfusion: Reconstruction and tracking of non-rigid scenes in real-time. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 343–352.
- Jeong Joon Park, Peter Florence, Julian Straub, Richard Newcombe, and Steven Lovegrove. 2019. DeepSDF: Learning continuous signed distance functions for shape representation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 165–174.
- Leonid I Rudin, Stanley Osher, and Emad Fatemi. 1992. Nonlinear total variation based noise removal algorithms. *Physica D: nonlinear phenomena* 60, 1–4 (1992), 259–268.
- Robert W Sumner, Johannes Schmid, and Mark Pauly. 2007. Embedded deformation for shape manipulation. In *ACM siggraph 2007 papers*. 80–es.
- Jianhua Wu and Leif P. Kobbelt. 2002. Fast Mesh Decimation by Multiple-Choice Techniques. In *International Symposium on Vision, Modeling, and Visualization*.
- Tao Yu, Zerong Zheng, Kaiwen Guo, Pengpeng Liu, Qionghai Dai, and Yebin Liu. 2021. Function4D: Real-time Human Volumetric Capture from Very Sparse Consumer RGBD Sensors. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 5746–5756.